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Cognitive trajectories in early Parkinson's disease: A multimodal MRI study of brain structure and cognitive decline

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ABSTRACT

Cognitive trajectories in Parkinson's disease (PD) exhibit substantial heterogeneity, ranging from long-term stability to rapid decline. The specific neural circuit dysfunctions underlying these distinct outcomes remain unclear. Utilizing magnetic resonance imaging (MRI) data from the Parkinson's Progression Markers Initiative cohort, we employed voxel-based morphometry and free-water imaging to analyze baseline and longitudinal changes in gray matter volume (GMV) and white matter microstructure over a four-year period in cognitively normal (PD-NC) and cognitively impaired (PD-CI) PD patients. Machine learning models were then used to integrate clinical, neuroimaging, and fluid biomarker data to explore factors associated with cognitive decline. At baseline, PD-NC patients exhibited larger GMV in six cerebellar regions compared to healthy controls. Longitudinally, PD-CI patients showed accelerated GMV loss in seven regions, including the thalamus and temporal lobes. Notably, baseline cerebellar GMV and longitudinal thalamic atrophy rates emerged as factors associated with steeper cognitive decline. Exploratory free-water analysis revealed that PD-CI patients exhibited altered macro-microstructural relationships in the thalamus. These findings implicate the cerebello–thalamo–cortical (CTC) circuit in the heterogeneity of cognitive decline in PD, with cerebellar structure and thalamic atrophy showing differential associations with cognitive trajectories. They further underscore the value of integrating longitudinal structural and microstructural measures to characterize disease-related cognitive changes.

1. Introduction

Cognitive impairment is a highly prevalent non-motor symptom in Parkinson's disease (PD) with a detrimental effect on the patients' quality of life (Aarsland et al., 2021). Approximately 20% of individuals newly diagnosed with PD present with mild cognitive impairment (MCI). Previous studies have revealed substantial heterogeneity in the cognitive trajectories of PD patients. Among patients with normal cognition at baseline, some maintain cognitive stability for years, while others progress rapidly to MCI or even dementia (Domellöf et al., 2015; Kawabata et al., 2024; Lawson et al., 2017; Pedersen et al., 2017; Weintraub et al., 2015). Exploring factors associated with inter-individual differences in the pattern and rate of cognitive decline, and characterizing associated brain structural alterations, may inform efforts toward more precise clinical management and prognostic evaluation.

The heterogeneity of cognitive trajectories in PD likely arises from multiple interacting pathological processes. Beyond well-established contributors such as Alzheimer's disease co-pathology (Aβ and tau),

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dopaminergic degeneration, neuroinflammation, and vascular changes (González-Redondo et al., 2014; Wang et al., 2023), emerging frameworks have proposed that the spatiotemporal propagation patterns of α-synuclein pathology may also contribute to clinical heterogeneity. For instance, the α-Synuclein Origin and Connectome (SOC) model hypothesizes distinct subtypes—the "body-first" and "brain-first"—with potentially different rates of cognitive progression (Borghammer, 2021). These diverse pathological processes may converge on shared downstream structural consequences, including macrostructural atrophy and microstructural tissue disintegration through neuronal loss and synaptic dysfunction (González-Redondo et al., 2014; Wang et al., 2023). Structural magnetic resonance imaging (MRI) provides valuable in vivo insights into such changes; however, prior findings demonstrated substantial heterogeneity. Cross-sectional studies have frequently demonstrated cortical atrophy in the frontal, parietal, temporal, and limbic regions in PD with cognitive impairment (PD-CI) compared to PD with normal cognition (PD-NC) (Gasca-Salas et al., 2019; Gorges et al., 2020; Mak et al., 2015, 2017; Segura et al., 2014). In contrast, other reports have found no significant changes or identified atrophy only in dementia patients (Agosta et al., 2014), collectively suggesting that static imaging may lack sensitivity in detecting early pathological changes, highlighting the need for longitudinal assessments to capture progressive neurodegeneration.

Longitudinal studies have shown that in early PD-MCI, there was a reduction in the volumes of subcortical nuclei (e.g., the hippocampus, thalamus, and amygdala), and that cortical atrophy emerged as MCI progressed to dementia (Foo et al., 2017; Hanganu et al., 2014; Mak et al., 2017; Xu et al., 2020), which is suggestive of cortical degeneration associated with disease advancement. However, existing longitudinal studies are often limited by short follow-up durations (typically of less than two years) and small sample sizes (Foo et al., 2017; Mak et al., 2017; Uribe et al., 2018; Xu et al., 2020), which limits their capacity to capture the continuous neurodegenerative progression from normal cognition to cognitive impairment. Most studies examining cognitive outcomes, including those employing machine learning models with multimodal data, tend to rely predominantly on baseline static measures and primarily focus on binary outcomes (conversion versus non-conversion) (Gasca-Salas et al., 2019; Shin et al., 2021). Although these approaches can identify individuals at higher risk (e.g., angular gyrus atrophy associated with conversion from PD-MCI to dementia) (Shin et al., 2021), they have limited ability to characterize the dynamic trajectory of cognitive decline or domain-specific changes, partly due to the predominant focus on baseline predictors.

White matter microstructural damage is regarded as a putative early imaging marker of PD-CI. Cross-sectional diffusion tensor imaging (DTI) studies have revealed widespread reductions in white matter integrity across the fronto-parietal and temporal tracts, as well as the corpus callosum, in PD-MCI, even in the absence of significant gray matter volume (GMV) atrophy (Agosta et al., 2014). A longitudinal study revealed that increased mean diffusivity in the left frontal cortex during early PD was associated with executive function decline (Sampedro et al., 2019). Furthermore, elevated thalamic diffusivity was found to be associated with overall cognitive decline (Zhang et al., 2016). However, the relationship between macrostructural changes and microstructural diffusion characteristics (He et al., 2024), particularly in the context of dynamic cognitive changes in PD, remains insufficiently investigated. Therefore, there is a need for multimodal analytical frameworks that integrate clinical data, baseline structural features, and longitudinal brain changes, while also incorporating microstructural measures, to more comprehensively characterize the cross-scale neurodegenerative processes underlying the heterogeneity of cognitive impairment in PD.

Given these research gaps, this study employed the Parkinson's Progression Markers Initiative (PPMI) cohort to investigate differences in baseline GMV as well as the longitudinal trajectories of GMV and cognitive function among PD patients with distinct cognitive outcomes. By integrating clinical features, fluid biomarkers, and longitudinal neuroimaging measures, we developed machine learning models to explore factors associated with cognitive change trajectories, thereby examining relationships between GMV alterations and concurrent cognitive decline. Additionally, free-water imaging analysis was employed to investigate associations between GMV alterations and underlying microstructural changes.

2. Methods

2.1. Participants

Longitudinal data were obtained from the PPMI database (https://www.ppmi-info.org). PPMI is an ongoing, comprehensive, international, multicenter observational study designed to identify biomarkers of PD progression (Marek et al., 2011). The study included de novo PD participants who met the following criteria: (1) normal baseline cognition, defined as an education-adjusted Montreal Cognitive Assessment (MoCA) score $\geq$ 26 (Nasreddine et al., 2005); (2) availability of complete cognitive assessment data at baseline and at the 4-year follow-up; and (3) availability of T1-weighted MRI scans at baseline and at least one subsequent follow-up time point (Year 1, Year 2, or Year 4). Participants were excluded if they were subsequently diagnosed with a non-PD condition during follow-up or if their MRI scans were deemed of inadequate quality.

Participants were categorized according to their cognitive status at the 4-year follow-up, based on the 'cogstate' variable provided in the PPMI dataset. According to the PPMI data dictionary (https://www.ppmi-info.org/access-data-specimens/download-data) and the PPMI Data User Guide (Section 11) (https://www.ppmi-info.org/), the cogstate variable represents an 'Investigator Diagnosis of Cognitive State' with values defined as: 1 = normal cognition, 2 = mild cognitive impairment (MCI), and 3 = dementia. The diagnosis is made annually by the site investigator based on neuropsychological test performance, functional independence (Schwab and England ADL scale), and clinical judgment (PPMI Data User Guide, Section 11) (Emre et al., 2007; Litvan et al., 2012).Participants with cogstate = 2 or 3 were classified as the cognitively impaired group (PD-CI). We acknowledge that the neuropsychological battery used in PPMI does not fully align with the MDS Level-II criteria, which recommend at least two tests per cognitive domain (Litvan et al., 2012).Nevertheless, the approach of combining MCI and dementia into a single cognitively impaired group has been used in prior PPMI-based studies (Schrag et al., 2017; Chen et al., 2023; Meng et al., 2023).

Data from HCs was also retrieved from the PPMI database. The inclusion criteria for HCs were: (1) normal baseline cognition; (2) availability of complete baseline cognitive assessment data; and (3) high-quality baseline T1-weighted MRI scans. A total of 133 HCs were included after matching to the PD cohort on age, sex, and years of education. Matching was performed at the group level using frequency matching.

The PPMI study protocol was approved by the institutional review boards at all participating sites, and written informed consent was obtained from all participants. The data used in this analysis were downloaded from the PPMI database on March 21, 2025, in compliance with the PPMI Data Use Agreement.

2.2. Data acquisition and assessment methods

2.2.1. Data availability and handling of missing data

Due to the prospective, multicenter design of the PPMI study, the availability of invasive biomarker and genetic data was contingent upon participant consent, sample availability, and technical success. Consequently, these data were not available for the entire cohort. Sample sizes for each biomarker assay at baseline and at the 4-year follow-up are provided in Table S1. In brief, cerebrospinal fluid (CSF) biomarkers (Aβ42, T-tau, P-tau, α-synuclein) exhibited the highest degree of

missingness, whereas serum biomarkers and most genetic data were largely complete. Baseline demographic and clinical comparisons presented in Table 1 are based on the available data for each variable. The specific strategies used to handle missing data in regression analyses are described in Section 2.4.

2.2.2. Clinical assessments

Global cognition was evaluated using the MoCA (Nasreddine et al., 2005). Specific cognitive domains were evaluated using a battery of validated tests (Benton, 1978; Gladsjo et al., 1999; Shapiro et al., 1999). Memory was assessed using the Hopkins Verbal Learning Test-Delayed Recall (HVLTDR) and Hopkins Verbal Learning Test-Recognition (HVLTREC). It should be noted that these two measures are derived from the same instrument and therefore do not constitute two independent tests per domain as recommended by MDS Level-II criteria (Litvan et al., 2012). Visuospatial function was evaluated with the Judgment of Line Orientation (JLO) (Benton, 1978). Processing speed and attention were assessed using the Symbol Digit Modalities Test (SDMT). Working memory and executive function were assessed using Letter-Number Sequencing (LNS) and the Semantic Fluency Test (SFT, animal category) (Gladsjo et al., 1999).

Motor symptoms were assessed using Parts II and III of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) and the Hoehn & Yahr (H&Y) stage. Motor subtypes were classified into tremor-dominant (TD), postural instability/gait difficulty (PIGD), or indeterminate subtypes (Stebbins et al., 2013).

A comprehensive set of validated scales was employed to evaluate other non-motor symptoms: Neuropsychiatric symptoms were assessed using the Geriatric Depression Scale (GDS) and the State-Trait Anxiety Inventory (STAI). Rapid eye movement sleep behavior disorder (RBD) was screened with the RBD Screening Questionnaire (RBDSQ); a score of $\geq$6 indicated probable RBD (Stiasny-Kolster et al., 2007). Daytime sleepiness was evaluated via the Epworth Sleepiness Scale (ESS). Impulse control disorders were assessed using the Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease (QUIP). Autonomic dysfunction was quantified by the Scales for Outcomes in PD-Autonomic (SCOPA-AUT). Orthostatic hypotension (OH) was defined as a sustained reduction in systolic blood pressure of $\geq$20 mmHg or diastolic blood pressure of $\geq$10 mmHg within 3 min of standing (Norcliffe-Kaufmann et al., 2018). Global non-motor burden was assessed using MDS-UPDRS Part I. Functional independence was assessed using the modified Schwab and England Activities of Daily Living (MSE-ADL) scale. Olfaction was evaluated using the University of Pennsylvania Smell Identification Test (UPSIT), with an age- and gender-adjusted score at or below the 15th percentile considered indicative of olfactory impairment (Siderowf et al., 2012).

2.2.3. Biomarker and genetic data

Fluid biomarker data included CSF concentrations of Aβ42, T-tau, P-tau, α-synuclein, and Neurofilament light chain (NFL), as well as serum concentrations of uric acid and NFL. Genetic variables included the number of APOE-ε4 alleles (0–2) and the mutation status of the LRRK2, GBA, and PRKN genes.

2.3. MRI data acquisition and analysis

High-resolution 3D T1-weighted images were acquired across PPMI sites using 3.0-Tesla MRI scanners (e.g., Siemens TrioTim) with a magnetization-prepared rapid acquisition gradient echo (MPRAGE) sequence. Key parameters were: slice thickness = 1.0–1.3 mm, repetition time (TR) = 2.3 s, echo time (TE) = 2.5–3.6 ms, flip angle = 9°, inversion time (TI) = 900 ms, matrix size = 240 × 256, and voxel size = 1.0 × 1.0 × 1.0 mm^3. DTI were acquired using a cardiac-gated spin-echo echo-planar imaging sequence with the following parameters: 64 diffusion directions, b-values = 0 and 1000 s/mm^2, TR = 760 ms, TE = 88 ms, flip angle = 90°, field of view = 224 × 224 mm, matrix size = 116 × 116, 72 contiguous axial slices, slice thickness = 2 mm, and a GRAPPA acceleration factor of 2.

Voxel-based morphometry (VBM) was performed using the Computational Anatomy Toolbox (CAT12, version 12.8.2, https://neuro-jena.github.io/cat/) within SPM12 (Statistical Parametric Mapping, https://www.fil.ion.ucl.ac.uk/spm/), implemented in MATLAB R2022b. The preprocessing pipeline included bias-field correction, tissue segmentation, spatial normalization to Montreal Neurological Institute (MNI) space, and nonlinear modulation (Jenkinson et al., 2012; Smith et al., 2004). To extract regional GMV, the Automated Anatomical Labeling (AAL) atlas comprising 90 cerebral and 26 cerebellar regions was applied. Total intracranial volume (TIV) was derived for use as a covariate in subsequent analyses. All preprocessed images underwent quality control procedures.

DTI data were preprocessed using FSL (version 6.0.4; https://fsl.fmrib.ox.ac.uk/fsl)(Jenkinson et al., 2012; Smith et al., 2004), with preprocessing steps including correction for eddy currents and head motion using eddy_correct (Andersson and Sotiropoulos, 2016) and brain extraction. To provide a more specific assessment of tissue microstructure, a two-compartment model was employed to separate the diffusion signal arising from tissue and that from extracellular free water (Arpin et al., 2022; Burciu et al., 2017; Planetta et al., 2016). This approach yielded two metrics: the Free-water fraction (FWf) and the Free-water-corrected fractional anisotropy (FAt)(Ofori et al., 2015; Pasternak et al., 2009). FWf may reflect increased extracellular space due to neuroinflammation, edema, brain atrophy, or neurodegenerative processes. FAt provides a more accurate measure of microstructural integrity by removing the confounding influence of free water. Finally, FWf and FAt values were extracted from the AAL atlas for quantitative analysis.

It should be noted that the diffusion MRI data were acquired with a single-shell protocol (b = 1000 s/mm^2), which may introduce bias in the estimation of FWf and FAt due to model assumptions. Additionally, the preprocessing did not include phase-encoding based distortion correction, which might affect regional accuracy, particularly in susceptibility-prone areas. These limitations are considered when interpreting the microstructural findings.

2.4. Statistical analysis

Baseline characteristics among HCs, PD-NC, and PD-CI were compared using one-way ANOVA for continuous variables and χ^2 tests for categorical variables, with Bonferroni-adjusted post hoc comparisons. Comparisons between PD-NC and PD-CI at baseline and at the 4-year follow-up were performed using *t*-tests or χ^2 tests.

To address multiple comparisons, we applied correction strategies tailored to each analysis. For primary analyses which directly tested the core hypotheses of this study—specifically, baseline GMV differences between groups and longitudinal GMV trajectories—we used Bonferroni correction to strictly control family-wise error rate. For secondary and sensitivity analyses, including: (1) sensitivity analyses of GMV incorporating Time × Age interactions; (2) cognitive trajectory analyses across multiple domains; and (3) free-water imaging analyses (which were limited by technical constraints including single-shell DTI acquisition and lack of distortion correction), we applied false discovery rate (FDR) correction using the Benjamini–Hochberg procedure (q = 0.05). These exploratory or hypothesis-generating analyses prioritized sensitivity to potential signals over strict control of Type I error.

Baseline GMV in brain regions defined by the AAL atlas was compared using Analysis of covariance (ANCOVA), with age, sex, education, and TIV included as covariates. To control for Type I error, *P* values were corrected for multiple comparisons using the Bonferroni method. Post-hoc pairwise comparisons were determined using a Bonferroni-corrected threshold.

Longitudinal changes in GMV were assessed using linear mixed-effects models (LMMs) with a random intercept for each subject. Fixed

Table 1
Baseline demographic, clinical, and biomarker characteristics of the study groups, and 4-year follow-up data for pd cognitive subgroups.

Variables	PD-NC(n = 97)		PD-CI(n = 36)		HCs(n = 133)	*p*	
	Baseline	Follow-up(4y)	Baseline	Follow-up(4y)	Baseline	Baseline	Follow-up(4y)
Age, years	58.29(9.69)	–	65.29(7.84)	–	59.70(11.65)	**0.003***,[a,c]	–
Male, n (%)	59(60.80)	–	25(69.40)	–	84(63.20)	0.658†	–
Education, years	15.81(3.01)	–	15.42(2.87)	–	16.11(2.72)	0.395*	–
Age at onset, years	53.89(13.49)	–	63.16(7.91)	–	–	**<0.001**	–
Disease duration, months	10.96(13.79)	–	8.14(9.06)	–	–	0.173	–
BMI	26.84(4.42)	26.61(4.34)	27.68(4.59)	27.03(3.82)	27.21(4.23)	0.594*	0.603
LEDD, mg	43.92(149.01)	546.77(323.21)	11.95(53.94)	516.04(248.94)	ns	0.072	0.607
MSEADLG, scores	94.43(5.72)	89.74(7.44)	94.72(5.34)	83.33(11.08)	99.96(0.43)	**<0.001***,[a,b]	**0.002**
MoCA	28.21(1.39)	28.3(1.65)	28.33(1.17)	24.53(4.66)	28.01(1.42)	0.346*	**<0.001**
JLO	12.62 (2.73)	12.61(2.73)	11.07(3.28)	11.26(3.32)	12.51(2.87)	**0.016***,[a,c]	**0.018**
HVLTDR	9.28(2.21)	9.14(2.52)	8.11(2.86)	6.81(3.62)	9.12(2.44)	**0.042***,[a]	**0.001**
HVLTREC	11.24(1.85)	11.4(0.90)	11.11(1.06)	11.06(1.29)	11.50(0.81)	0.171*	0.143
LNS	12.22(2.58)	12.11(2.43)	10.75(2.84)	9.47(3.57)	12.20(2.68)	**0.010**[a,c]	**<0.001**
SDMT	46.71 (9.13)	47.20(7.92)	43.07(10.00)	37.62(11.17)	51.13(10.58)	**<0.001***,[b,c]	**<0.001**
SFT	11.36 (2.67)	11.51(2.88)	9.69(2.60)	9.39(3.48)	11.54(3.22)	**0.004***,[a,c]	**0.001**
Part I	4.64(3.85)	7.55(4.95)	6.00(4.50)	13.14(7.95)	3.32(2.97)	**<0.001***,[b,c]	**<0.001**
Part II	5.64(3.96)	8.58(5.55)	5.06(3.85)	13.33(7.72)	0.50(0.96)	**<0.001***,[b,c]	**<0.001**
Part III	19.59(9.26)	26.89(13.99)	24.50(8.31)	32.64(14.13)	1.77(2.29)	**<0.001***,[a,b,c]	**0.038**
H&Y Stage 2–3, n (%)	15(15.50)	44.88(27.15)	12(33.30)	60.22(32.02)	–	**0.015**†	**0.028**†
TD	76(78.4)	54(55.7)	26(72.2)	16(44.4)	–		
Indeterminate	10(10.30)	14(14.4)	2(5.6)	5(13.9)	–	0.227†	0.420†
PIGD	11(11.3)	29(29.9)	8(22.2)	15(41.7)	–		
GDS, scores	5.22(1.56)	2.21(2.59)	4.92(2.10)	3.61(3.04)	1.20(2.08)	**<0.001***,[b,c]	**0.009**
STAAR, scores	92.30(9.55)	62.12(17.53)	87.83(12.64)	72.72(22.09)	53.39(12.37)	**<0.001***,[b,c]	**0.005**
Hallucinations, scores	0.03(0.174)	0.12(0.42)	0.08(0.28)	0.5(0.91)	0(0)	**0.009***,[c]	**0.022**
Apathy, scores	0.14(0.5)	0.26(0.62)	0.22(0.485)	0.86(1.10)	0.06(0.269)	0.061*	**0.003**
DDS, scores	0.02(0.143)	0.04(0.20)	0.03(0.167)	0.17(0.51)	0.04(0.227)	0.802*	0.157
QUIP, scores	3.34(1.72)	0.4(0.79)	2.67(2.32)	0.58(0.99)	0.27(0.76)	**<0.001***,[b,c]	0.275
SCOPA-AUT, scores	8.36(5.89)	11.36(5.45)	11.78(7.26)	16.51(8.51)	6.02(4.27)	**<0.001***,[a,b,c]	**0.002**
OH, n (%)	9(9.30)	12(12.40)	4(11.10)	5(13.90)	1(0.80)	**0.004**†,[b,c]	0.777†
hyposmia, n (%)	66(68.00)	–	31(86.10)	–	12(9.00)	**<0.001**†,[b,c]	–
Constipation, scores	0.31(0.601)	0.57(0.71)	0.5(0.737)	0.81(0.75)	0.14(0.351)	**0.001***,[b,c]	0.091
Probable RBD, n (%)	10(10.30)	30(30.9)	10(27.80)	19(52.8)	9(6.80)	**0.002**†,[a,c]	**0.020**†
RBDSQ, scores	2.18(2.18)	4.4(3.281)	3.78(2.71)	6.44(3.836)	2.41(2.25)	**0.002***,[a,c]	**0.003**
ESS, scores	6.37(3.16)	7.47(4.682)	8.44(4.17)	10.44(5.959)	5.17(3.29)	**<0.001***,[a,b,c]	**0.003**
APOE-e4 alleles (0/1/2)	0.32(0.55)	–	0.31(0.58)	–	0.23(0.49)	0.435*	–
LRRK2 carrier, n (%)	8(8.20)	–	1(2.80)	–	0(0.00)	**0.003**†,[b]	–
GBA carrier, n (%)	10(10.30)	–	2(5.60)	–	0(0.00)	**0.001**†,[c]	–
PRKN carrier, n (%)	3(3.10)	–	0(0.00)	–	1(0.80)	0.266†	–
CSF Biospecimen, pg/ml							
$A\beta_{42}$	889.65(352.79)	798.94(301.75)	732.99(304.09)	626.46(257.55)	–	**0.022**	**0.019**
T-tau	161.35(51.34)	157.61(58.18)	157.37(59.86)	159.66(87.34)	–	0.709	0.899
P-tau	13.60(4.94)	32.31(146.87)	13.15(5.44)	95.85(350.47)	–	0.656	0.252
α-synuclein	1492.41(671.77)	–	1300.99(489.66)	–	–	0.138	–
Nfl	91.72(47.80)	–	89.29(30.14)	–	–	0.847	–
Serum Biospecimen, pg/ml							
uric acid	5.29(1.44)	5.07(1.80)	5.54(1.20)	5.02(1.31)	–	0.367	0.877
Nfl	11.24(5.28)	–	12.81(6.12)	–	–	0.167	–

Data are presented as mean ± standard deviation (SD) for continuous variables, and as frequency (n) with percentage (%) for categorical variables.
Group comparisons were conducted using ANOVA for continuous variables (indicated with *) and the χ^2 test for categorical variables (indicated with †).
Independent-samples t-tests were additionally performed between PD-NC and PD-CI groups.
Significant p-values ($p < 0.05$) are shown in bold.
Abbreviation: PD-NC, Parkinson's disease with normal cognition; PD-CI, Parkinson's disease with cognitive impairment; HCs, healthy controls; SD, standard deviation; BMI, body mass index; LEDD, levodopa equivalent daily dose; MSEADL, Modified Schwab & England Activities of Daily Living Score; MoCA, Montreal Cognitive Assessment; JLO, Judgment of Line Orientation; HVLTDR, Hopkins Verbal Learning Test Delayed Recall; HVLTREC, Hopkins Verbal Learning Test Delayed Recognition; LNS, Letter-Number Sequencing Test; SDMT, Symbol Digit Modalities Test; SFT, Semantic Fluency total animal number; MDS-UPDRS I-III, Movement Disorder Society Unified Parkinson's Disease Rating Scale parts I-III; H&Y, Hoehn and Yahr stage; TD, Tremor-Dominant; PIGD, Postural Instability/Gait Difficulty; GDS, Geriatric Depression Scale total score; STAI, State-Trait Anxiety Index total score; DDS, Dopamine Dysregulation Syndrome; QUIP, Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease Score; SCOPA-AUT, Scales for Outcomes in Parkinson's disease for Autonomic Symptoms; OH, orthostatic hypotension; RBD, Rapid Eye Movement Sleep Behavior Disorder; ESS, Epworth Sleepiness Scale Score; APOE-e4, number of E4 alleles in APOE genotype; LRRK2, Leucine-Rich Repeat Kinase 2 gene; GBA, β-glucocerebrosidase gene; PRKN, Parkin RBR E3 Ubiquitin Protein Ligase gene; CSF, cerebrospinal fluid; NfL, neurofilament light chain; T-tau, total tau protein; P-tau, phosphorylated tau protein.
[a] Significant post hoc comparison (Bonferroni-corrected *p* value<0.05) for PD-NC vs. PD-CI.
[b] Significant post hoc comparison (Bonferroni-corrected *p* value<0.05) for PD-NC vs. HC.
[c] Significant post hoc comparison (Bonferroni-corrected *p* value <0.05) for PD-CI vs. HC.

effects were group, time, and their interaction. Covariates included baseline age, sex, education, and TIV. For each ROI, the primary linear mixed model was specified as:

$$Volume_{\{ij\}} = \beta_{\{0\}} + \beta_{\{1\}Time_{\{ij\}}} + \beta_{\{2\}Group_{\{i\}}} + \beta_{\{3\}(Time_{\{ij\}} \times Group_{\{i\}})} + \beta_{\{4\}Education_{\{i\}}} + \beta_{\{5\}Sex_{\{i\}}} + \beta_{\{6\}Age_{\{i\}}} + \beta_{\{7\}TIV_{\{i\}}} + b_{\{0i\}} + \varepsilon_{\{ij\}}$$

where $Volume_{\{ij\}}$ denotes gray matter volume for subject at visit; $\beta_{\{0\}}$ is the fixed intercept; $\beta_{\{1\}}$ estimates the annual rate of GMV change in the PD-NC group (reference category); $\beta_{\{2\}}$ captures the baseline GMV difference between the PD-CI and PD-NC groups; $\beta_{\{3\}}$ is the key term of interest, representing the difference in the annual rate of GMV change between PD-CI and PD-NC; $\beta_{\{4\}}$–$\beta_{\{7\}}$ correspond to covariates (years of education, sex, baseline age, TIV); $b_{\{0i\}}$is the subject-specific random intercept; and $\varepsilon_{\{ij\}}$ denotes the residual error term. The Bonferroni correction was used to control for multiple comparisons.

Furthermore, given the close association between age and brain volume alterations, we conducted a sensitivity analysis to assess the robustness of the main analysis results. Specifically, we additionally included a Time × Age interaction term in the LMMs to account for the potential influence of age on the longitudinal rate of GMV change. The FDR method was employed to control for multiple comparisons. All mixed-effects models were implemented using the lme4 package in R.

To model longitudinal trajectories of cognitive performance, we constructed LMMs separately for MoCA, HVLTDR, LNS, SDMT, JLO, and SFT, including a random intercept for each subject. The fixed effects were group, time, and their interaction. Covariates included baseline age, sex, and education. Significant Group × Time interactions were interpreted as evidence of differential trajectories of cognitive decline between the PD-NC and PD-CI groups. Sensitivity analyses were conducted with the Time × Age interaction term included as an additional fixed effect. The *P* values from both the main and sensitivity analyses were adjusted for multiple comparisons using the FDR method.

To identify multimodal factors associated with decline in specific cognitive domains, we constructed multiple linear regression models to examine factors associated with longitudinal slopes of cognitive tests that exhibited a significant Time × Group interaction in the preceding LMMs analysis (i.e., MoCA, HVLTDR, LNS, and SDMT slopes). Models were constructed separately for the PD-NC and PD-CI groups, using four distinct sets of predictors: (1) Clinical and Demographic Features: baseline variables that differed significantly between the PD-CI and PD-NC groups; (2) Baseline Brain Morphometry: baseline GMV of ROIs exhibiting significant group differences in the baseline ANCOVA analysis; (3) Longitudinal Brain Atrophy: GMV slopes from ROIs that showed a significant Time × Group interaction in the LMMs; and (4) Integrated Multimodal: a combination of all features from the preceding three sets.

Predictor selection was performed using stepwise regression based on the Akaike Information Criterion (AIC), which balances model fit with parsimony and has been suggested as a reasonable approach in exploratory settings with limited sample sizes when accompanied by rigorous validation (Heinze and Dunkler, 2017). Model generalizability was assessed using leave-one-out cross-validation (LOOCV). Model performance was evaluated using the Pearson correlation coefficient (r) between the LOOCV-predicted and observed slopes (with statistical significance defined as $\alpha < 0.05$), as well as R^2, adjusted R^2, Root mean square error (RMSE), and Mean absolute error (MAE).

To ensure robustness, we conducted extensive sensitivity analyses. For the primary analyses, mean imputation within group was applied to facilitate model fitting. Given the limitations of mean imputation, we additionally performed multiple imputation using the mice package in R (20 imputed datasets), repeating stepwise regression in each imputed dataset to evaluate predictor selection consistency. Multicollinearity was assessed using variance inflation factor (VIF), with values >5 indicating moderate collinearity. For variables with VIF > 5, we conducted sensitivity analyses by refitting models with one of the correlated variables excluded to evaluate coefficient stability. All analyses were conducted in R using the tidyverse, boot, mice, and ggpubr packages.

To further investigate the relationship between regional GMV and microstructural alterations, for regions that exhibited significant inter-group differences in GMV in both baseline cross-sectional and longitudinal analyses, FAt and FWf were compared across the three groups, with age, sex, and education included as covariates. To account for multiple comparisons, a FDR correction was applied to each region using the Benjamini–Hochberg procedure. Longitudinal changes in FAt and FWf were assessed using LMMs, following the same analytical approach as that used for GMV. Sensitivity analyses were also conducted.

Macro-microstructural associations were assessed using two approaches. Cross-sectionally, partial correlations between GMV and FWf were computed controlling for age, sex, education, and TIV. Longitudinally, Pearson correlations were calculated between GMV and FWf slopes. Inter-group differences in correlation strengths were tested using Fisher's z-transformation, with FDR correction applied to all *p*-values.

3. Result

Fig. 1 provides a schematic illustration of the study design and the multimodal framework developed to characterize cognitive decline in de novo PD. This longitudinal study recruited 133 de novo PD patients with normal cognition and 133 matched healthy controls (HCs) from the PPMI database. Clinical and imaging data were collected at baseline and at annual follow-up visits (Years 1, 2, and 4). At the Year 4 time point, PD participants were categorized as either cognitively normal (PD-NC, $n = 97$) or cognitively impaired (PD-CI, $n = 36$) based on cognitive assessments (see Supplementary Figs. S1 and S2). The multimodal data preprocessing pipeline integrated clinical data, biofluid markers, and longitudinal neuroimaging metrics. The neuroimaging data included GMV from T1-weighted MRI, as well as FWf and FAt derived from DTI. The analytical framework combined cross-sectional and longitudinal neuroimaging data to examine baseline GMV differences and group differences in atrophy rates. These processed metrics were subsequently integrated for three analytical purposes: (i) to assess macro–microstructural associations by examining relationships between GMV and white matter integrity indices; (ii) to identify key factors associated with cognitive decline using stepwise regression analysis; and (iii) to evaluate model generalizability in explaining inter-individual variability in longitudinal cognitive trajectories using LOOCV.

3.1. Baseline demographic, clinical, and biomarker characteristics

The comprehensive baseline and 4-year follow-up characteristics for all study groups are presented in Table 1. At baseline, the PD-CI group was significantly older than the PD-NC and HCs groups ($p = 0.003$). Sex, education level, and body mass index did not differ significantly between groups. In contrast, PD-CI patients demonstrated more severe motor symptoms, characterized by higher MDS-UPDRS Part III scores and a greater proportion at H&Y stage 2–3 ($p = 0.015$).

Cognitive assessment revealed that, despite comparable performance on the MoCA and HVLTREC, the PD-CI group showed significant impairments on specific tests, including JLO, LNS, and SFT, with scores lower than those of both PD-NC and HCs participants (all $p < 0.05$). PD-CI also demonstrated significantly lower HVLTDR scores compared with those of PD-NC ($p < 0.05$). SDMT scores were significantly higher in HCs participants relative to both PD groups ($p < 0.001$), and no statistically significant difference was observed between PD-NC and PD-CI groups.

SCOPA-AUT and ESS scores differed across groups, with mean values lowest in HCs, intermediate in PD-NC, and highest in PD-CI (all $p < 0.05$). The prevalence of probable RBD and RBDSQ scores were significantly higher among PD-CI participants than among both PD-NC and HCs (all $p < 0.05$). Depression, anxiety, constipation, impulse control disorders, and OH were more severe in both PD groups relative to HCs (all $p < 0.05$), but no significant differences were observed between the

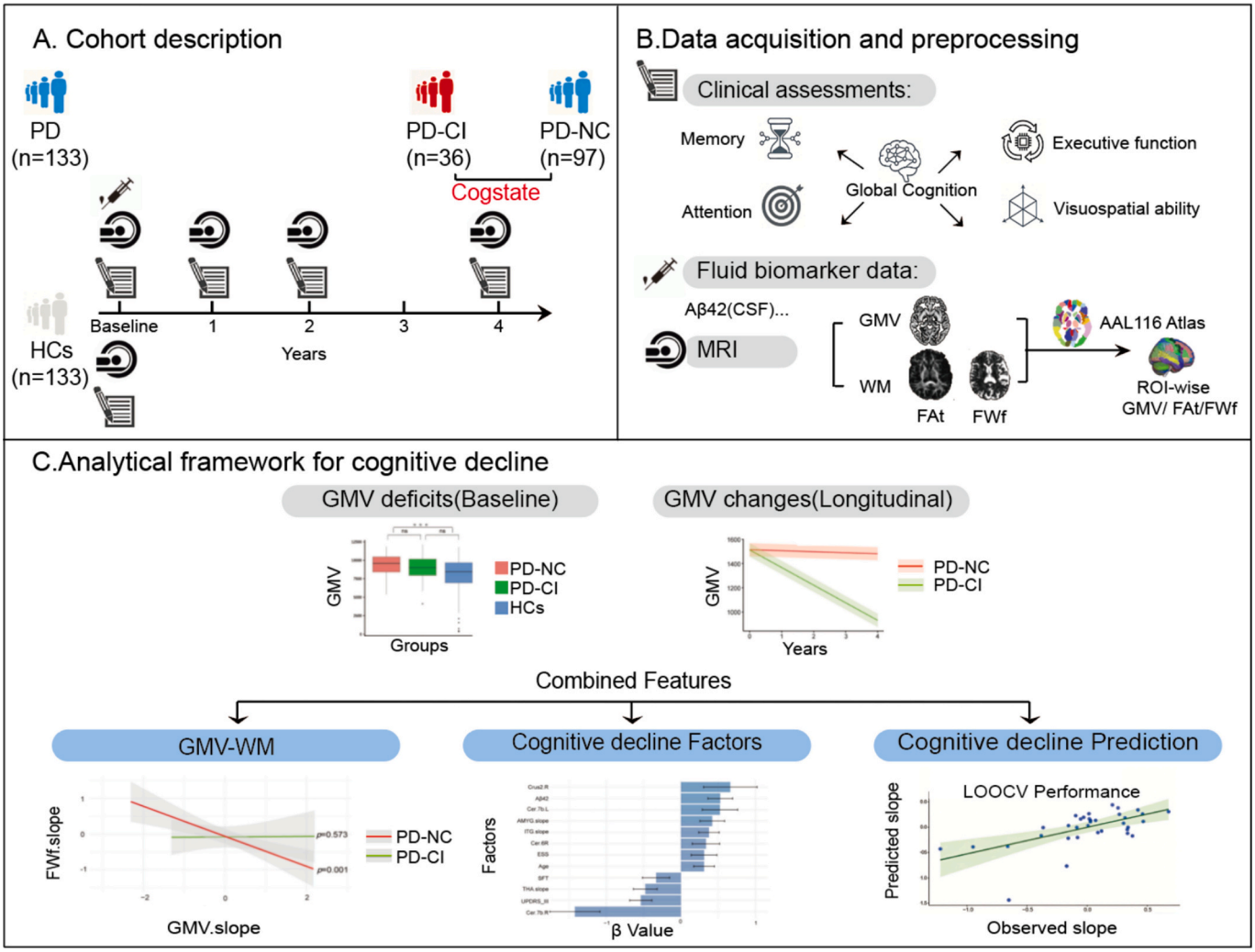


Fig. 1. Study design and multimodal analytical framework. (A) Cohort description: 133 de novo PD patients with normal cognition and 133 age-, sex-, and education-matched HCs from PPMI. Clinical, biofluid, and imaging data acquired at baseline and years 1, 2, and 4. At year 4, PD participants classified as cognitively normal (PD-NC, $n = 97$) or impaired (PD-CI, $n = 36$) based on PPMI cogstate variable. (B) Data preprocessing: Regional GMV extracted from T1-weighted MRI using CAT12 and AAL-116 atlas. DTI-derived free-water fraction (FWf) and free-water-corrected FA (FAt) from FSL. (C) Analytical framework integrated cross-sectional and longitudinal imaging with clinical data to: (i) examine macro–microstructural associations; (ii) identify factors associated with cognitive decline via stepwise regression; (iii) evaluate model explanatory power using LOOCV.

PD subgroups. Hallucinations and apathy were more severe in PD-CI participants than in HCs (all $p < 0.05$), but did not significantly differ between PD subgroups.

From the perspective of fluid biomarkers, CSF Aβ42 levels were significantly lower in PD-CI participants than in PD-NC participants ($p = 0.022$). No significant inter-group differences were observed in CSF T-tau, P-tau, α-synuclein, or serum/CSF NFL. Genetically, the prevalence of LRRK2 mutation carriers was higher in PD-NC participants than in HCs ($p = 0.003$), whereas the prevalence of GBA mutations was greater in PD-CI participants than in HCs ($p = 0.001$).

3.2. Four-year clinical and biomarker progression

At the 4-year follow-up, PD-CI participants showed more severe clinical symptoms compared with PD-NC participants. Motor symptoms were more severe in PD-CI participants, as evidenced by higher MDS-UPDRS Part II and III scores and a greater proportion with H&Y stage ≥2 (all $p < 0.05$), while motor subtype distribution was not significantly different between groups.

Regarding cognition, PD-CI participants performed worse than PD-NC participants on the MoCA, JLO, HVLTDR, LNS, SDMT, and SFT (all $p < 0.05$), whereas no significant difference was found for HVLTREC ($p = 0.143$).

For non-motor symptoms, PD-CI participants had higher scores on the MDS-UPDRS Part I, GDS, STAI, hallucinations, apathy, SCOPA-AUT, RBDSQ, and ESS (all $p < 0.05$). The prevalence of probable RBD remained higher in PD-CI participants ($p = 0.020$).

With respect to fluid biomarkers, consistent with the baseline findings, CSF Aβ42 levels were lower in PD-CI participants than in PD-NC participants ($p = 0.019$). No significant inter-group differences were observed for CSF T-tau, P-tau, or serum uric acid.

3.3. Cross-sectional gray matter volume differences

ANCOVA revealed significant cross-sectional GMV differences following Bonferroni correction for multiple comparisons, which were primarily localized to cerebellar regions (see Fig. 2, Table S2). Specifically, GMV in bilateral Crus II (Crus2.L/R) exhibited significant inter-group differences ($F = 14.417$, pBonf <0.001; $F = 13.020$, pBonf <0.001). Significant differences were also observed in the right cerebellar lobule VI (Cer.6.R), bilateral lobule VIIb (Cer.7b.L/R), and right lobule VIII (Cer.8.R), all with pBonf <0.05. Post-hoc analysis with

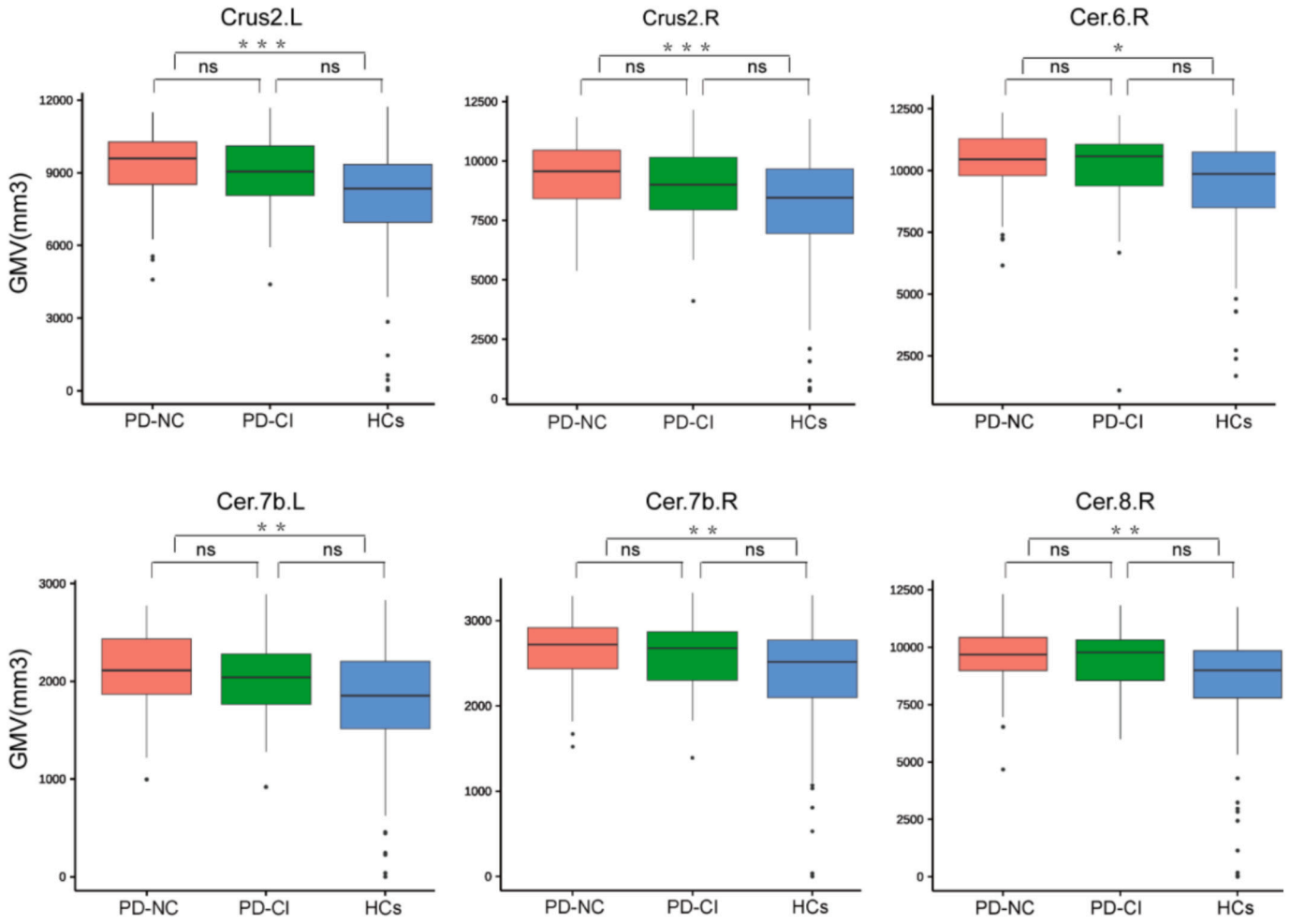


Fig. 2. Cross-sectional group differences in baseline gray matter volume. ANCOVA results comparing GMV across HCs (n = 133), PD-NC (n = 97), and PD-CI (n = 36), adjusted for age, sex, education, and TIV. Post-hoc pairwise comparisons with Bonferroni correction. Asterisks indicate significance levels (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$); ns, not significant. Data presented as estimated marginal means ±95% CI. See Table S2 for full statistics. Abbreviations: PD-NC, Parkinson's disease with normal cognition; PD-CI, Parkinson's disease with cognitive impairment; HCs, healthy controls; GMV, gray matter volume; L, left; R, right; Cer, cerebellar lobule; Crus, cerebellar crus.

Bonferroni-adjusted thresholds indicated that PD-NC had larger GMV in these regions than HCs, whereas PD-CI exhibited intermediate GMV values with no statistically significant differences from either PD-NC or HCs. No cerebral cortical or subcortical regions remained significant after Bonferroni correction.

3.4. Longitudinal trajectories of gray matter volume

LMMs identified seven cerebral regions exhibiting significantly accelerated GMV loss in the PD-CI group over the four-year period, as illustrated in Fig. 3A-B and Table S3: the left amygdala (AMYG.L), bilateral fusiform gyri (FFG.L/R), right thalamus (THA.R), left temporal pole of the middle temporal gyrus (TPOmid.L), and bilateral inferior temporal gyri (ITG.L/R). All regions exhibited significant Time × Group interactions after Bonferroni correction (pBonf <0.05) with negative estimates, indicating a faster rate of GMV atrophy in the PD-CI group compared to the PD-NC group. The largest effect sizes, reflected by the most negative estimates, were observed in the right thalamus (estimate = −136.97 mm^3/year, 95% CI [−190.77 mm^3/year, −83.17], pBonf <0.001) and the right inferior temporal gyrus (estimate = −136.71 mm^3/year, 95% CI [−199.87, −73.55], pBonf = 0.006). Sensitivity analyses incorporating Time × Age interactions, with FDR correction for multiple comparisons, confirmed the robustness of these findings (see Table S4).

3.5. Longitudinal cognitive trajectories

Compared with the PD-NC group, the PD-CI group demonstrated a steeper decline in global cognition (MoCA, β = −0.70 per year, 95% CI [−0.95, −0.45], pFDR <0.001), verbal memory (HVLTDR, β = −0.41 per year, 95% CI [−0.64, −0.17], pFDR = 0.002), working memory/executive function (LNS, β = −0.34 per year, 95% CI [−0.59, −0.09], pFDR = 0.013), and processing speed (SDMT, β = −1.56 per year, 95% CI [−2.41, −0.70],pFDR = 0.001) (Fig. 3C, Table S5). No significant inter-group differences in longitudinal decline were observed for visuospatial function (JLO, pFDR>0.05) or semantic fluency (SFT, pFDR >0.05). Sensitivity analyses incorporating Time × Age interactions, with FDR correction, confirmed that the accelerated cognitive decline in PD-CI was largely independent of age (see Fig. 3C and Table S5). In these sensitivity models, the estimates for HVLTDR and LNS were slightly attenuated and reached borderline significance after FDR correction (pFDR = 0.06), yet their uncorrected 95% confidence intervals did not include zero.

3.6. Explanatory modeling of cognitive decline based on brain structure

3.6.1. Model performance comparison

To systematically evaluate the ability of multimodal indicators to model variance in cognitive decline, we developed 32 multiple linear

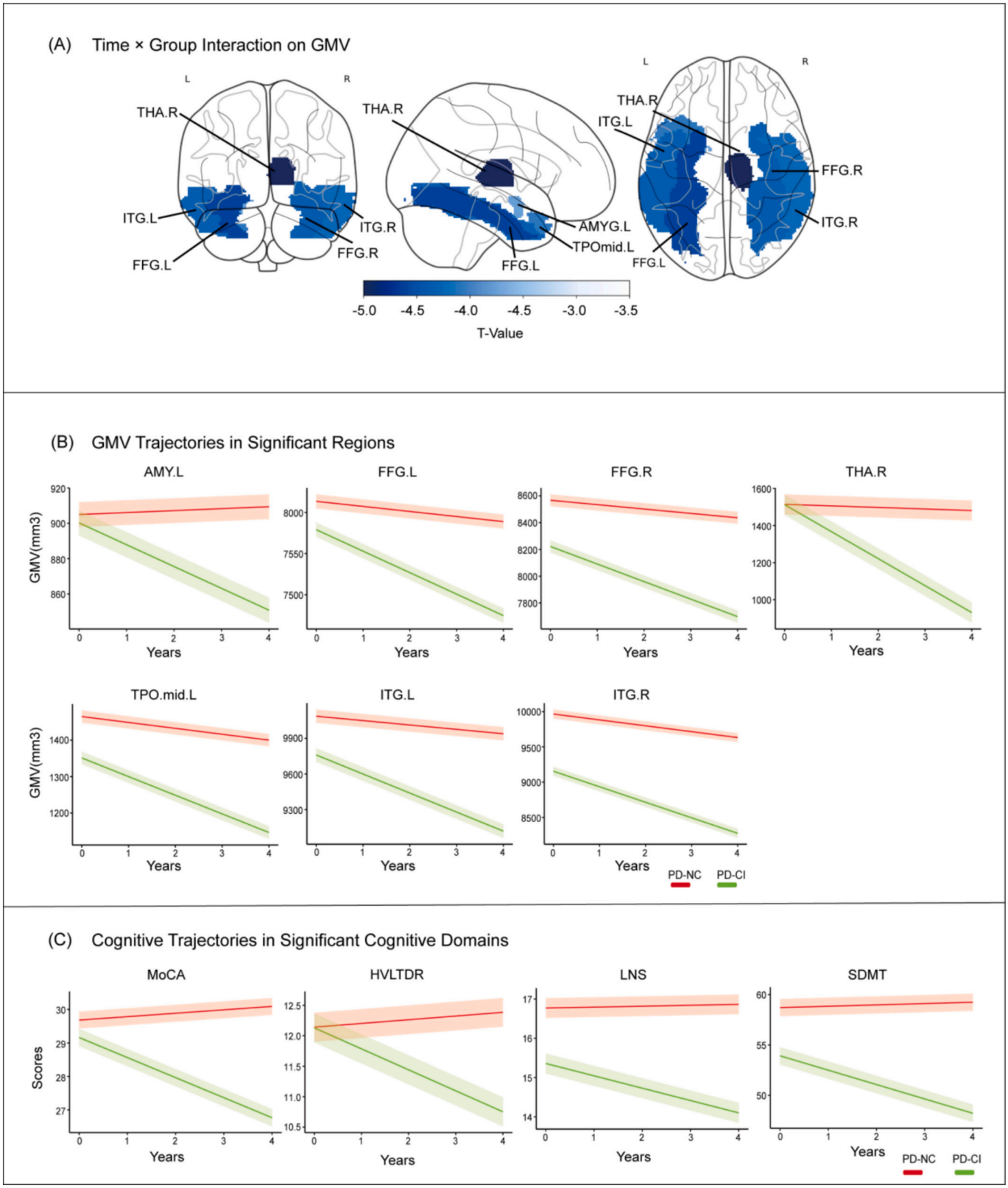


Fig. 3. Time × Group interaction effects on GMV and cognitive function in PD-CI and PD-NC groups. (A) Statistical map depicting brain regions exhibiting a significant Time × Group interaction in GMV (Bonferroni-corrected $p < 0.05$), adjusted for age, sex, education, and TIV. The color bar represents t-values; darker colors indicate higher absolute t-values, reflecting greater GMV atrophy in the PD-CI group. (B) Model-fitted GMV trajectories for regions in (A). Red, PD-NC (n = 97); green, PD-CI (n = 36). Shaded areas, 95% CI. (C) Model-fitted cognitive trajectories (LMMs, FDR-corrected $p < 0.05$), adjusted for age, sex, education, Shaded areas, 95% CI. Abbreviations: AMYG, amygdala; FFG, fusiform gyrus; THA, thalamus; TPOmid, temporal pole mid-temporal gyrus; ITG, inferior temporal gyrus; L, left; R, right; MoCA, Montreal Cognitive Assessment; HVLTDR, Hopkins Verbal Learning Test–Delayed Recall; LNS, Letter-Number Sequencing; SDMT, Symbol Digit Modalities Test. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

regression models for the PD-CI and PD-NC groups separately. These models targeted the decline slopes of global cognition (MoCA slope) and three cognitive domains: working memory (LNS slope), information processing speed (SDMT slope), and delayed recall (HVLTDR slope). Predictor selection was performed using stepwise regression, and model generalizability was assessed using LOOCV. Performance metrics for these models are summarized in Table S6.

Overall, for most cognitive domains in both the PD-CI and PD-NC groups, the proportion of model generalizability (assessed by LOOCV r) by the integrated multimodal models was consistently greater than that of any single-modality model (Fig. 4A). For instance, in the PD-CI group, the integrated multimodal model for the LNS slope (LOOCV $r = 0.53$) showed higher LOOCV r values than the best single-modality model (baseline brain morphology model, LOOCV $r = 0.40$). Furthermore, the models developed for the PD-CI group yielded higher LOOCV r values than those for the PD-NC group (Fig. 4B). In the PD-CI group, the integrated multimodal models for MoCA, LNS, HVLTDR, and SDMT slopes all showed statistically significant explanatory power, with LOOCV correlation coefficients ranging from 0.46 to 0.53 (all $p \leq 0.005$). In contrast, within the PD-NC group, only the HVLTDR slope model achieved statistical significance (LOOCV $r = 0.45$, $p < 0.001$).

3.6.2. *Predictor analysis of the optimal model*

Across the multiple integrated multimodal models in the PD-CI group, several variables were retained across multiple domain-specific models (Tables S7–S8, Fig. 4C). The best-performing model was the LNS slope model in the PD-CI group, which incorporated 12 candidate variables retained after stepwise selection (Fig. 4D). Variables

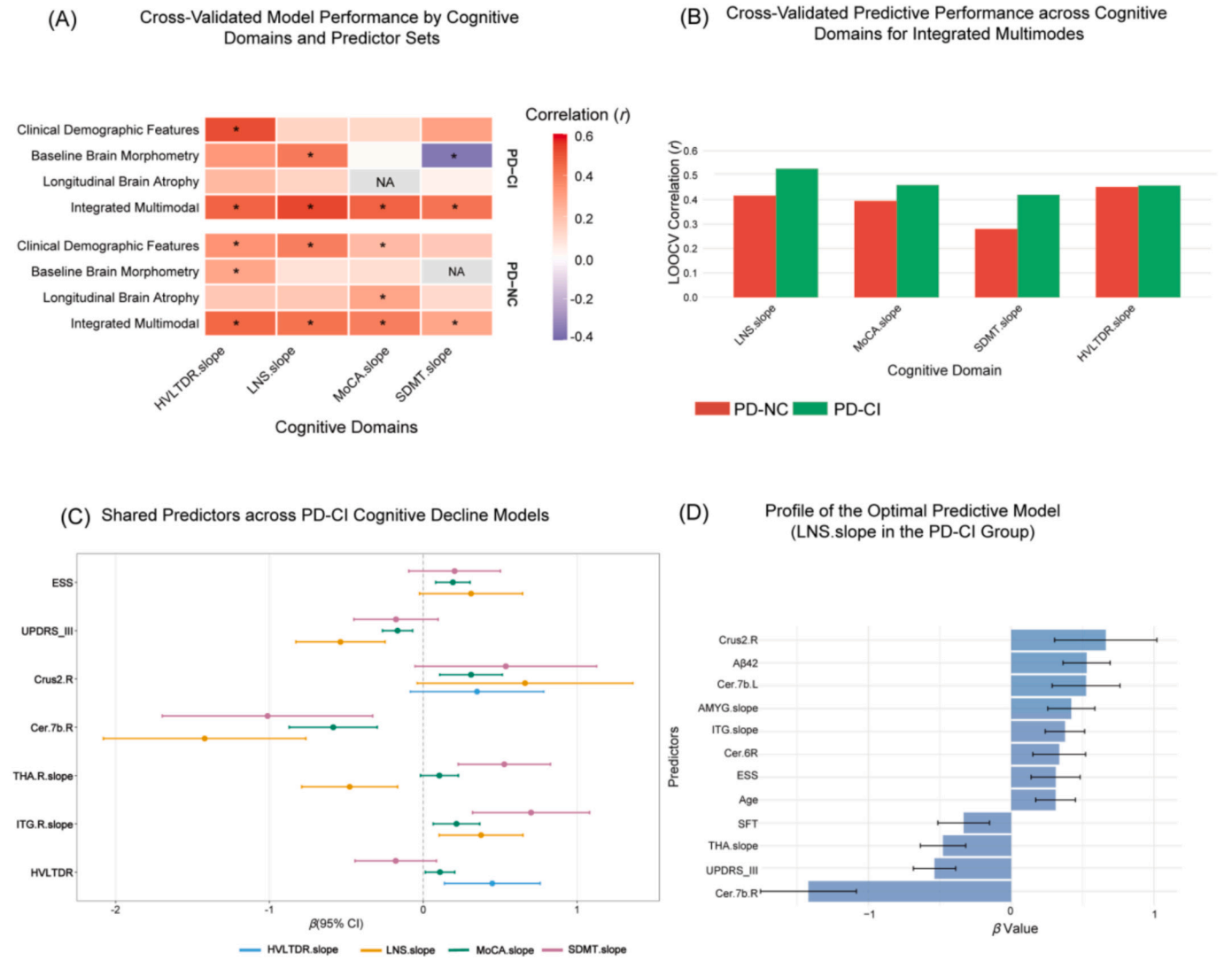


Fig. 4. Explanatory performance of multimodal models and factors associated with cognitive decline in PD subgroups. (A) Heatmap of LOOCV correlation coefficients (r) for all 32 models. Color scale indicates strength of association between predicted and observed slopes. Asterisks indicate $p < 0.05$. Two models (MoCA slope in PD-CI; SDMT slope in PD-NC) excluded due to no features meeting stepwise selection criteria (NA). (B) LOOCV performance (r) comparison between PD-NC (red) and PD-CI (green) for integrated multimodal models across four cognitive domains. (C) Forest plot of factors associated with cognitive decline in ≥2 optimal models (LNS, HVLTDR, MoCA slopes) in PD-CI. Points, standardized β coefficients; horizontal lines, 95% CI. Colors indicate respective models. (D) Standardized β coefficients for all variables retained in the optimal LNS slope model for PD-CI (n = 36). Negative coefficients: MDS-UPDRS III ($\beta = -0.538$, $p = 0.001$), Cer.7b.R ($\beta = -1.421$, $p < 0.001$), THA.R.slope ($\beta = -0.478$, $p = 0.006$). Positive coefficients: Aβ ($\beta = 0.527$, $p = 0.004$), AMYG.L.slope ($\beta = 0.420$, $p = 0.018$), Cer.7b.L ($\beta = 0.524$, $p = 0.038$), age ($\beta = 0.310$, $p = 0.036$), ITG.R.slope ($\beta = 0.376$, $p = 0.012$). See Tables S6–S11 for full statistics and sensitivity analyses. Abbreviations: HVLTDR, Hopkins Verbal Learning Test–Delayed Recall; LNS, Letter-Number Sequencing; MoCA, Montreal Cognitive Assessment; SDMT, Symbol Digit Modalities Test; LOOCV, leave-one-out cross-validation; ESS, Epworth Sleepiness Scale; UPDRS-III, Unified Parkinson's Disease Rating Scale part III; THA.R.slope, atrophy rate of right thalamus; ITG.R.slope, atrophy rate of right inferior temporal gyrus; AMYG.L.slope, atrophy rate of left amygdala; SFT, semantic fluency total animal number. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

negatively associated with LNS slope included baseline MDS-UPDRS Part III score ($\beta = -0.538$, $p = 0.001$), baseline GMV of the right cerebellar lobule VIIb (Cer.7b.R; $\beta = -1.421$, $p < 0.001$), and the atrophy slope of the right thalamus ($\beta = -0.478$, $p = 0.006$). Variables positively associated with LNS slope (i.e., associated with slower decline) included cerebrospinal fluid Aβ42 concentration ($\beta = 0.527$, $p = 0.004$), the atrophy slope of the left amygdala ($\beta = 0.420$, $p = 0.018$), baseline GMV of the left cerebellar lobule VIIb ($\beta = 0.524$, $p = 0.038$), age ($\beta = 0.310$, $p = 0.036$), and the atrophy slope of the right inferior temporal gyrus (ITG.R; $\beta = 0.376$, $p = 0.012$).

To evaluate the robustness of this model, we assessed multicollinearity using VIF. Most variables showed VIF values below 5, with the exception of Crus2. R (VIF = 9.28) and Cer.7b.R (VIF = 8.09), and the Pearson correlation between these variables was $r = 0.89$. We therefore re-estimated the model after sequentially excluding Crus2. R or Cer.7b.R (Table S10). Aβ, Cer.7b.L, and THA.R.slope remained statistically significant with consistent effect directions across all model specifications. LOOCV performance remained similar across models ($r = 0.49$–0.55, Table S11). Multiple imputation analysis yielded consistent variable selection across 20 imputed datasets (Table S9).

Beyond the LNS model, several variables showed consistent associations across multiple cognitive domains in the PD-CI group. Baseline MDS-UPDRS Part III showed negative regression coefficients in the LNS ($\beta = -0.54$), MoCA ($\beta = -0.17$), and SDMT ($\beta = -0.18$) slope models. Baseline GMV of the right cerebellar Crus II (Crus2.R) was positively associated with cognitive slopes across domains: LNS ($\beta = 0.66$), HVLTDR ($\beta = 0.35$), MoCA ($\beta = 0.31$), and SDMT ($\beta = 0.54$). The coefficient for Crus2. R showed variability across model specifications. In contrast, baseline GMV of the right cerebellar lobule VIIb (Cer.7b.R) was negatively associated with cognitive slopes across domains: LNS ($\beta = -1.42$), MoCA ($\beta = -0.59$), and SDMT ($\beta = -1.01$). This association was consistently observed across multiple imputation (selected in all 20 datasets, mean $p < 0.001$) and alternative model specifications (Tables S9–S11). In the MoCA slope model of the PD-NC subgroup, Cer.7b.R showed a positive association ($\beta = 0.10$, $p = 0.027$) (Table S8).

3.7. Associations between microstructural and macrostructural measures

After quality control, baseline DTI data were obtained from 115 PD patients for cross-sectional analysis. For longitudinal slope estimation, data from 127 PD patients who underwent at least two DTI scans during follow-up visits were included. Some patients lacked a baseline DTI scan but provided two follow-up scans, thereby increasing the sample size for slope analysis. It should be noted that the DTI data were acquired with a single-shell protocol and preprocessing did not include phase-encoding based distortion correction; therefore, the following findings should be considered exploratory.

Exploratory DTI analysis examined microstructural measures in regions identified in the GMV analyses. Cross-sectional analysis of baseline microstructural measures revealed a significant intergroup difference in FWf of the right cerebellar lobule VI (Cer.6.R; $p < 0.001$, pFDR = 0.004). Post-hoc analysis revealed a gradient of increasing FWf values from PD-NC (0.278 ± 0.004) to PD-CI (0.284 ± 0.007) and HCs (0.287 ± 0.005), though pairwise comparisons did not remain significant after FDR correction. A similar trend was also observed in the left cerebellar lobule VIIb (Cer.7b.L; $p = 0.036$, pFDR = 0.208). No significant baseline differences were detected in other cerebellar subregions for either FAt or FWf measures (see Fig. 5A and Table S12).

Longitudinally, the right thalamus exhibited accelerated microstructural changes in PD-CI compared to the PD-NC group, characterized by an accelerated decline in FAt and a concurrent increase in FWf (significant Group × Time interactions, both pFDR <0.05). A significant Group × Time interaction was also observed for FAt in the Cer.6.R region in PD-CI (pFDR <0.016) (see Fig. 5C and Table S13).

Next, we explored the relationship between microstructural and macrostructural measures. Cross-sectionally, FWf and GMV in the Cer.6.R showed a negative correlation in HCs ($r = -0.451$, pFDR = 0.021) and a trend-level negative correlation in PD-NC ($r = -0.292$, pFDR = 0.054). This association was not observed in the PD-CI group ($r = -0.214$, pFDR = 0.305) (Fig. 5B, Table S14).

Longitudinal slope analysis revealed region-specific differences in correlations between microstructural and macrostructural change rates.: in the Cer.6.R region, The relationship between FAt and GMV slopes in the Cer.6.R region showed an uncorrected group difference ($p = 0.022$), characterized by a positive correlation in the PD-NC group and a negative correlation in the PD-CI group. However, this difference did not remain significant after FDR correction (pFDR = 0.088) (see Fig. 5E and Table S15). Conversely, in the PD-NC group, right thalamic FWf slope was negatively correlated with GMV slope ($r = -0.335$, pFDR = 0.008). This association was not observed in the PD-CI group ($r = -0.112$, $p = 0.482$) (Fig. 5D, Table S15).

4. Discussion

This study demonstrates that a multimodal framework integrating baseline clinical features, baseline brain structure, and longitudinal brain structural change rates can explain a substantial portion of variance in domain-specific cognitive trajectories among early PD patients. Further analyses suggest that alterations in the cerebello-thalamo-cortical (CTC) circuit may be associated with different cognitive trajectories: higher baseline cerebellar GMV was associated with slower cognitive decline, while accelerated atrophy in the thalamus and temporal regions was linked to more rapid deterioration. These findings provide multimodal imaging evidence that may help clarify the neural substrates underlying cognitive heterogeneity in PD. Among these findings, thalamic atrophy and longitudinal cognitive trajectories represent the most robust results, as they consistently survived multiple comparison correction and were supported across analytical approaches. Baseline cerebellar GMV differences, while also statistically significant, should be interpreted as supportive findings that may provide complementary context for understanding cognitive heterogeneity.

Baseline analysis revealed that cognitively stable patients (PD-NC) had larger GMV in multiple cerebellar regions, particularly Crus II, compared to healthy controls, while the PD-CI group showed intermediate values. In regression models, baseline GMV of the right Crus II was consistently associated with slower cognitive decline across domains. Exploratory free-water imaging revealed a pattern of group differences in cerebellar FWf, with numerically lower values in the PD-NC group. A trend-level negative correlation between FWf and GMV was observed in the PD-NC group, directionally consistent with the significant correlation seen in HCs; this association was not evident in the PD-CI group. Together, these findings are consistent with the concept of cerebellar reserve—the capacity to recruit alternative neural resources to maintain function despite pathology (Mitoma et al., 2020), though this interpretation remains hypothesis-generating and requires validation in independent cohorts.

In contrast, the right cerebellar lobule VIIb (Cer.7b.R) displayed a markedly different pattern. In the PD-CI group, its baseline GMV showed a strong negative association with cognitive slopes—an association that remained robust across multiple imputation and extensive sensitivity analyses. In the PD-NC group, however, this region showed a weak positive association with MoCA slope. This bidirectional association suggests that Cer.7b.R may play context-dependent roles depending on disease trajectory. The high correlation between adjacent cerebellar regions in our dataset precludes definitive attribution of opposing effects to individual lobules; nonetheless, the overall pattern aligns with recent studies implicating the cerebellum as a potential hub of structural vulnerability in subsets of PD patients (Duanmu et al., 2025), highlighting the functional heterogeneity of cerebellar regions in PD.

Longitudinal analysis identified accelerated atrophy in seven brain regions in the PD-CI group, with the right thalamus showing the largest group difference in annualized atrophy rate—an annualized atrophy

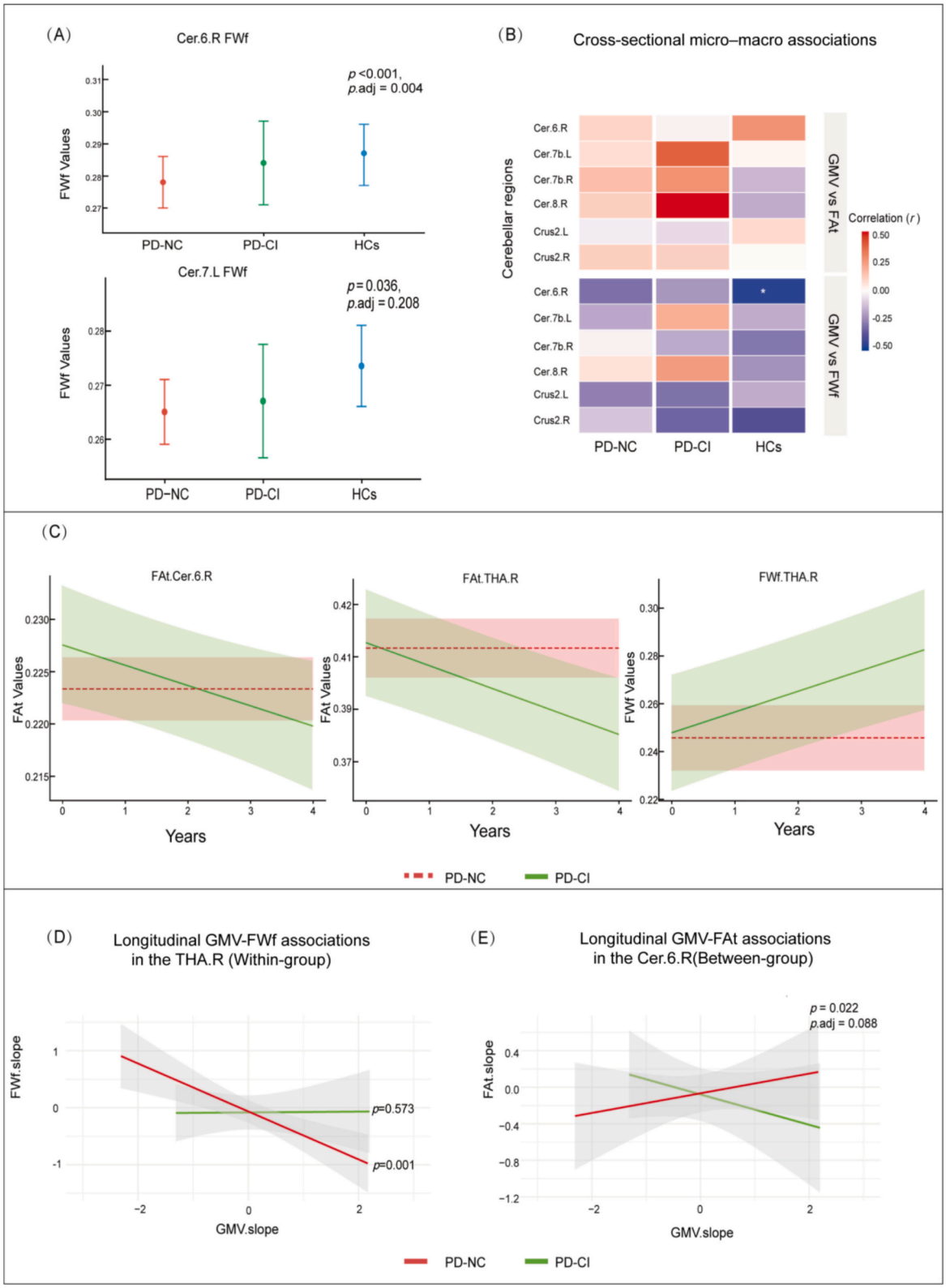


Fig. 5. Cross-sectional and longitudinal cerebellar and thalamic microstructure. (A) Baseline FWf comparisons in cerebellar subregions. ANCOVA adjusted for age, sex, education. Right lobule VI showed significant group difference (pFDR = 0.004). Data as estimated marginal means ±95% CI. (B) Cross-sectional partial correlations between microstructural metrics (FAt, FWf) and GMV in cerebellar regions, adjusted for age, sex, education, TIV. Color scale indicates correlation strength. (C) Longitudinal microstructural trajectories from LMMs showing significant Time × Group interactions. Right thalamus: PD-CI showed faster FAt decline and FWf increase (pFDR <0.05). Right lobule VI: PD-CI showed faster FAt decline (pFDR <0.05). Shaded areas, 95% CI. (D) Within-group correlations between longitudinal GMV and FWf slopes in right thalamus. PD-NC: $r = -0.335$, pFDR = 0.008; PD-CI: $r = -0.112$, $p = 0.482$. (E) Between-group comparison of GMV–FAt slope correlations in right lobule VI. Fisher's z-test $p = 0.022$ (pFDR = 0.088). Shaded areas, 95% CI. See Tables S9–S12 for full statistics. Abbreviations: PD-NC, Parkinson's disease with normal cognition; PD-CI, Parkinson's disease with cognitive impairment; FAt, free-water-corrected fractional anisotropy; FWf, free-water fraction; GMV, gray matter volume; DTI, diffusion tensor imaging; CI, confidence interval; TIV, total intracranial volume.

rate approximately 136 mm^3 faster than in PD-NC. Diffusion tensor imaging provided complementary insights into the nature of thalamic involvement: PD-CI patients exhibited accelerated decline in FAt and a concurrent increase in FWf, along with a loss of the coordinated macro–microstructural relationship observed in cognitively stable patients (where faster FWf increase correlated with more rapid GMV atrophy). This pattern suggests that thalamic pathology in declining patients extends beyond volumetric loss to encompass tissue microstructural alteration—a finding consistent with previous reports linking thalamic diffusivity changes to cognitive decline in PD (Zhang et al., 2016), faster thalamic atrophy was associated with slower working memory decline within the PD-CI group—an association that remained statistically robust across extensive sensitivity analyses. While counterintuitive, given the modest sample size, this finding should be interpreted cautiously. Possible explanations include statistical fluctuation given the modest sample size, survivor bias, or a dynamic process of network reorganization, though the latter remains speculative. Replication in larger cohorts is required before drawing mechanistic conclusions.

Among the models examined, the model for working memory/executive function (LNS) slope in the PD-CI group showed the strongest explanatory power, consistent with evidence that executive function is a particularly vulnerable domain in early PD cognitive decline (Young et al., 2024). The LNS slope was associated with multiple factors, with varying degrees of stability across sensitivity analyses. Baseline motor symptom severity (MDS-UPDRS III) showed a consistent negative association in most model specifications, suggesting a relationship between motor burden and executive dysfunction. Lower CSF Aβ42 levels were robustly associated with faster LNS decline across all model specifications, consistent with the "multiple hit" hypothesis (Citro et al., 2024). Notably, older age was associated with slower LNS decline within the PD-CI group—an association that was not consistently observed across all sensitivity analyses. This may reflect a floor effect (older patients having lower baseline scores, limiting detectable decline) or survivor bias (older individuals remaining in the cohort representing a slower-progressing subgroup). This age-related association should therefore be interpreted with caution.

The observed dissociation—cognitively stable patients showing preserved cerebellar structure and declining patients exhibiting accelerated thalamic and temporal atrophy—raises the possibility that differential patterns of structural vulnerability may contribute to cognitive heterogeneity. At the group level, the PD-CI cohort showed a higher prevalence of questionnaire-based probable RBD (27.8%) compared with PD-NC (approximately 10%). However, our data do not permit formal subtype classification or direct inference regarding propagation patterns. Questionnaire-based RBD assessment is not equivalent to polysomnography-confirmed RBD, and we lack α-synuclein seeding assays to directly test these hypotheses. Therefore, this interpretation remains speculative. Multiple mechanisms—including genetic factors such as GBA mutations (Pasternak et al., 2009), concomitant Alzheimer's disease pathology, and other processes—likely contribute to cognitive decline in this heterogeneous population.

4.1. Translational implications

If replicated in independent and adequately powered cohorts, the imaging features identified in this study—including baseline cerebellar GMV patterns, longitudinal thalamic atrophy rates, and associated microstructural alterations—may contribute to the development of multimodal risk stratification frameworks for cognitive decline in PD. The concept of cerebellar structural reserve, if supported by future mechanistic and longitudinal investigations, could provide a theoretical basis for exploring targeted interventions. For example, non-invasive neuromodulation approaches targeting cerebellar regions (e.g., Crus II) have shown cognitive effects in other populations (Yao et al., 2022). However, their relevance to PD-related cognitive impairment remains untested. Any translational application should therefore be regarded as exploratory and contingent upon replication, mechanistic validation, and rigorous clinical trials.

4.2. Limitations

Several limitations should be considered when interpreting these results. First, the PD-CI group sample size was relatively small ($n = 36$), increasing susceptibility to overfitting in multivariable analyses. Although LOOCV was applied to estimate generalizability and multiple sensitivity analyses (including multiple imputation and alternative model specifications) were conducted to assess robustness, independent replication remains essential. Second, the linear mixed-effects models included random intercepts but not random slopes, which may not fully capture inter-individual variability in longitudinal trajectories. Third, diffusion MRI was acquired using a single-shell protocol, which may limit the precision of free-water parameter estimation. In addition, preprocessing did not include phase-encoding-based distortion correction, which could affect regional accuracy in susceptibility-prone areas. Fourth, the cognitive assessment battery used in PPMI does not fully align with MDS Level-II criteria (Litvan et al., 2012), specifically, the memory domain was assessed using two subtests from the same instrument (HVLT recall and recognition), which MDS guidelines consider as a single test, and the PPMI cognitive battery does not include a dedicated language domain assessment. This suboptimal assessment limits the sensitivity and specificity of cognitive group classification and potentially constrains the comprehensiveness of cognitive characterization. Fifth, RBD status was determined using questionnaire-based criteria rather than polysomnography, precluding definitive diagnostic classification. Sixth, the absence of molecular pathological markers (e.g., α-synuclein seeding assays) and functional imaging limits the ability to directly link structural alterations to underlying circuit dysfunction.

4.3. Future directions

Future studies should validate these findings in larger, independent cohorts, incorporating polysomnography for definitive RBD assessment, α-synuclein seeding assays from skin or biofluids (Liu et al., 2025) genetic profiling (e.g., GBA, LRRK2), and functional imaging to directly link structural changes to circuit-level dysfunction. Data-driven disease progression modeling approaches (e.g., SuStaIn (Oh et al., 2024)) may further facilitate characterization of spatiotemporal heterogeneity at the individual level. Given the instability of some region-specific associations observed in our sensitivity analyses—particularly those involving highly collinear cerebellar subregions—replication in adequately powered cohorts is essential before any clinical translation can be considered. Ultimately, integrating structural, microstructural, and molecular biomarkers may enable precision medicine approaches that match individual patients with targeted interventions based on their specific pattern of neurodegeneration.

5. Conclusions

By integrating multimodal longitudinal data, this study suggests that cognitive heterogeneity in early Parkinson's disease is closely associated with both the baseline state and dynamic evolution of the cerebello–thalamo–cortical (CTC) circuit. Cognitively stable patients exhibited greater cerebellar volume at baseline, consistent with the concept of structural reserve as a hypothesis-generating observation, whereas patients who declined demonstrated accelerated atrophy in thalamic and temporal regions. Multimodal models incorporating both baseline and longitudinal structural measures explained more variance in cognitive trajectories than single-timepoint models alone. Together, these findings implicate the CTC circuit as a potentially important substrate underlying divergent cognitive outcomes in PD and indicate that longitudinal tracking of circuit integrity may improve characterization of risk for decline.

CRediT authorship contribution statement

Chunmei Wen: Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yi Xing:** Methodology, Investigation, Formal analysis, Data curation. **Hao Zhou:** Visualization, Methodology, Investigation, Formal analysis, Data curation. **Jingzhe Li:** Visualization, Software, Data curation. **Yukun Feng:** Software, Methodology, Data curation. **Yinhao Xu:** Methodology, Investigation, Data curation. **Haicun Shi:** Validation. **Xiao Hu:** Validation, Software, Investigation. **Junjie Zheng:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Weiguo Liu:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Ethics statement

The PPMI study protocol was approved by the institutional review boards at all participating sites.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors declare that no generative AI or AI-assisted technologies were used in the writing process of this manuscript.

Declaration of competing interest

All authors declare no financial or non-financial competing interests. All authors have duly completed the conflict-of-interest disclosure forms as stipulated by the International Committee of Medical Journal Editors (ICMJE), which are available upon request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nbd.2026.107407.

Data availability

All data of this study can be found in Parkinson Progression Markers Initiative (PPMI) database (www.ppmi-info.org).

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